Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Agentic AI for Market Research

TGS Ref No: TGS-2022017520

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 14.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
13.0	Legacy supplied deck	GA4 courseware baseline retained as coverage floor.	Tertiary Infotech Academy
14.0	13 September 2026	Agentic research rebuild:Claude Cowork marketing,Claude Code,14 Activities,source checks,GA4 measurement contracts and filed60+90 minute assessment timing.	Tertiary Infotech Academy

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Course Overview

Agentic AI for Market Research · TGS-2022017520 · ICT-SNM-3007-1.1. 16 hours across2 days:13.5 training plus2.5 assessment. Cowork marketing synthesis and Claude Code reproducible research use synthetic evidence with GA4-compatible measurement contracts.

Learning Outcomes

LO1: Identify market-research data types and evaluate sources for quality, relevance and ethical use.

LO2: Analyse market trends and variables using traceable evidence and reproducible methods.

LO3: Explore customer behaviour, segments and journeys and develop bounded forecasts.

LO4: Evaluate marketing effectiveness using measurement models, events and conversions.

Duration and Assessment Reconciliation

Current filed WA specifies60 minutes and PP 90 minutes. Use15:30–18:00 on Day 2. This preserves the actual TMS13.5+2.5 hour record despite the public page's2 hour assessment description. Day 1 has8 teaching hours;Day 2 has5.5 teaching hours. Each day has60 minutes lunch,excluded from teaching totals.

Daily Schedule

Day 1

Time	Minutes	Topic / Activity	Method	Slides
09:00–10:00	60	Activity 01: Evidence Registry	Mechanism exhibit;guided practice;review	14–26
10:00–11:00	60	Activity 02: Cowork Folder Research Brief	Mechanism exhibit;guided practice;review	27–39
11:00–12:15	75	Activity 03: Competitor Triangulation	Mechanism exhibit;guided practice;review	41–53
12:15–13:30	75	Activity 04: Review Themes and Bias	Mechanism exhibit;guided practice;review	54–66
13:30–14:30	60	Lunch	Break	—
14:30–15:45	75	Activity 05: Segments and Customer Value	Mechanism exhibit;guided practice;review	68–80
15:45–17:00	75	Activity 06: Journey and Funnel Diagnostics	Mechanism exhibit;guided practice;review	81–93
17:00–18:00	60	Activity 07: GA4 Measurement Contract	Mechanism exhibit;guided practice;review	95–107

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Day 2

Time	Minutes	Topic / Activity	Method	Slides
09:00–09:45	45	Activity 08: Event Quality and Deduplication	Mechanism exhibit;guided practice;review	108–120
09:45–10:30	45	Activity 09: Conversions and Attribution Limits	Mechanism exhibit;guided practice;review	121–133
10:30–11:15	45	Activity 10: Lifecycle Creative from Evidence	Mechanism exhibit;guided practice;review	135–147
11:15–12:00	45	Activity 11: SEO Content Experiment	Mechanism exhibit;guided practice;review	149–161
12:00–13:00	60	Lunch	Break	—
13:00–13:45	45	Activity 12: Bounded Demand Forecast	Mechanism exhibit;guided practice;review	163–175
13:45–14:45	60	Activity 13: Claude Code Skills and MCP Audit	Mechanism exhibit;guided practice;review	177–189
14:45–15:30	45	Activity 14: Effectiveness Decision Capstone	Mechanism exhibit;guided practice;review	191–203
15:30–16:30	60	Written Assessment K1–K8	Individual open book	—
16:30–18:00	90	Practical Performance A1–A8	Individual open book	—

Topic and Activity Breakdown

Activity 01 — Evidence Registry · Slides 14–26

LO1. Classify primary/secondary and qualitative/quantitative sources; record first-party origin, time window and consent; separate direct observations from interpretations.

Practice:activities/activity-01-evidence-registry · mock-data/sources.csv

Acceptance:Four uniquely identified sources; each claim states population, period, confidence and an explicit limitation.

Trainer diagnosis:A global report is used to claim Singapore buying intent. Its population and date do not match the decision, so the claim must be downgraded.

Activity 02 — Cowork Folder Research Brief · Slides 27–39

LO1. Inventory -> bounded plan -> approved file reads -> evidence synthesis -> output diff -> human decision. This is a teaching model of task control, not a fabricated Cowork interface.

Practice:activities/activity-02-cowork-folder-brief · mock-data/campaign-files.csv

Acceptance:Four manifest records; original input hashes unchanged; two requested outputs; untrusted instruction quoted as data and ignored.

Trainer diagnosis:An external note tells the agent to upload the folder elsewhere. It is task data, not an authorised instruction, and must not change scope.

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Activity 03 — Competitor Triangulation · Slides 41–53

LO2. Match comparable features, units and retrieval dates; distinguish vendor claim from independent evidence; triangulate one claim with two source types before promoting it.

Practice:activities/activity-03-competitor-triangulation · mock-data/competitors.csv

Acceptance:Hourly prices 60,45,50; all three claim IDs cited; at least two uncertainty statements; no unsupported leadership claim.

Trainer diagnosis:Cowork labels Harbour the market leader from a 4.8 rating based on eight reviews. Rating and evidence volume must be evaluated separately.

Activity 04 — Review Themes and Bias · Slides 54–66

LO2. Define codebook -> quote evidence -> assign theme -> review disagreement -> count -> qualify interpretation. Keep negation and mixed sentiment visible.

Practice:activities/activity-04-review-themes-bias · mock-data/reviews.csv

Acceptance:Eight review IDs retained; 3 timing,3 relevance,2 support; R08 mixed qualifier; selection-bias limitation.

Trainer diagnosis:R08 is flattened into positive sentiment although it says the exercises were rushed. Mixed sentiment needs a separate qualifier.

Activity 05 — Segments and Customer Value · Slides 68–80

LO3. Choose observable non-sensitive variables, compute group sizes and values, validate distinct needs with evidence, avoid assigning personas beyond available data.

Practice:activities/activity-05-segments-customer-value · mock-data/segments.csv

Acceptance:Margins/customer 300,200,500; total customers 80; no invented demographic traits; retained-margin assumptions explicit.

Trainer diagnosis:The agent invents an age range and income for each segment. Neither field exists; the persona must remain evidence-constrained.

Activity 06 — Journey and Funnel Diagnostics · Slides 81–93

LO3. Use journey-events.csv to validate session-level ordered events in one synthetic cohort and30-day window;derive the aggregate stage counts,then compute conditional and overall rates. Raw aggregate counts alone do not establish a funnel.

Practice:activities/activity-06-journey-funnel-diagnostics · mock-data/funnel.csv

Acceptance:Row-level journey-events.csv confirms2400 unique sessions,96 enquiries,48 qualified and12 enrolments;ordered stages share the same session IDs;rates 4%,50%,25%,0.5%;two alternate friction explanations.

Trainer diagnosis:A month-end enrolment count is divided by a different month’s enquiries. Time-window mismatch invalidates the conversion rate.

Activity 07 — GA4 Measurement Contract · Slides 95–107

LO4. Business question -> event trigger -> recommended event -> parameter schema -> consent -> validation evidence -> key-event designation. Avoid treating button clicks as successful submissions.

Practice:activities/activity-07-ga4-measurement-contract · mock-data/measurement-plan.csv

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Acceptance:Four events with unambiguous success triggers; purchase transaction_id,currency,value,items; no personal-data parameters.

Trainer diagnosis:generate_lead fires when a form button is pressed even if validation fails. The event trigger must be confirmed successful submission.

Activity 08 — Event Quality and Deduplication · Slides 108–120

LO4. Parse schema -> isolate purchases -> validate transaction IDs -> group duplicates -> reconcile value -> emit accepted and quarantined rows -> human review.

Practice:activities/activity-08-event-quality-deduplication · mock-data/events.csv

Acceptance:Five raw rows retained in audit; T001 counted once; T002 once; E04 missing-ID; accepted value $1100.

Trainer diagnosis:Deduplicating by value removes two different $600 orders. Transaction identity, not price equality, determines duplication.

Activity 09 — Conversions and Attribution Limits · Slides 121–133

LO4. Define eligible channels and lookback -> order touches -> choose allocation model -> reconcile totals -> compare sensitivity -> distinguish attribution from lift.

Practice:activities/activity-09-conversions-attribution-limits · mock-data/touchpoints.csv

Acceptance:Both models reconcile to $2400; paid first-touch $1200; email last-touch $1300; no causal ROI claim.

Trainer diagnosis:The agent cuts paid search because last-touch credit is zero. Assisted demand and incremental impact require separate investigation.

Activity 10 — Lifecycle Creative from Evidence · Slides 135–147

LO3. Evidence brief -> segment need -> approved claim -> channel/consent constraint -> draft variant -> human brand review -> measurement hypothesis.

Practice:activities/activity-10-lifecycle-creative-evidence · mock-data/lifecycle.csv

Acceptance:Three segment drafts; claim IDs and CTA; no invented scarcity; send_authorised false; rates use sent.

Trainer diagnosis:A draft says only two seats remain without inventory evidence. The urgency claim must be removed or verified by an authorised source.

Activity 11 — SEO Content Experiment · Slides 149–161

LO2. Normalise query -> classify intent -> map page/claim -> calculate CTR and enquiry rates -> isolate one content change -> define control/window -> assess uncertainty.

Practice:activities/activity-11-seo-content-experiment · mock-data/search-content.csv

Acceptance:CTRs 4%,3%,6%,3%; training enquiry/click 12.5%; one isolated change and explicit confounder.

Trainer diagnosis:A content plan targets the highest-impression query and ignores enquiry quality. Optimise against the decision metric rather than raw reach.

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Activity 12 — Bounded Demand Forecast · Slides 163–175

LO3. Check regular intervals -> reserve test period -> compute naive baseline -> compare moving average -> record error -> state scenario range -> monitor change.

Practice:activities/activity-12-bounded-demand-forecast · mock-data/demand.csv

Acceptance:Naive 104; moving-average 100; August moving-average error 10.67; campaign shift and seasonality caveats.

Trainer diagnosis:The agent fits an elaborate model to six points and reports a precise probability interval. The data do not support such precision.

Activity 13 — Claude Code Skills and MCP Audit · Slides 177–189

LO2. User question -> reviewed Skill -> approved read-only MCP tool -> schema validation -> deterministic calculation -> source-linked report -> reviewer. CSV fallback tests the same data contract.

Practice:activities/activity-13-claude-code-skills-mcp · mock-data/tool-results.csv

Acceptance:Four source IDs retained; rates 5%,4%,8%,1%,overall 4%; local MCP initialize/list/call test passes; Skill front matter valid; Claude invocation separately evidenced;no live account claimed.

Trainer diagnosis:The agent treats the Skill as permission to access a production account. Instruction reuse and tool authorisation are separate controls.

Activity 14 — Effectiveness Decision Capstone · Slides 191–203

LO4. Reconcile inputs -> compute denominator-aware metrics -> apply margin -> inspect attribution -> challenge confounders -> recommend bounded test -> human approve -> monitor.

Practice:activities/activity-14-effectiveness-decision-capstone · mock-data/campaigns.csv

Acceptance:96 leads; spend $2100; revenue $6000; contribution $900; paid CPL40 and contribution0; human approval pending.

Trainer diagnosis:The report calls paid search profitable because ROAS is 2. At a 50% gross margin, contribution after media spend is zero before other costs.

Supplementary Filed Coverage

Slide204: Behavioural Analytics and Primary Research — K1 · Combine behavioural traces with primary research; agent output is an analysis artifact, not a new source.

Slide205: Demographic Segmentation and Bias — K2 · Demographic and behavioural variables answer different questions; do not invent persona attributes.

Slide206: Pricing Evidence and Willingness to Pay — K3 · GA4 can describe behaviour around offers; independent pricing research and controlled tests are needed for demand interpretation.

Slide207: Survey Questions That Test a Decision — K4/A3 · Define decision,population,response options and sampling limits before agent-assisted coding.

Slide208: GA4 Account-to-Stream Boundary — Legacy setup coverage retained as architecture; exact click paths are in the LG and current official documentation.

Slide209: Dimensions, Metrics and Denominators — LO1/LO2 · A report export is interpretable only when its dimension,metric,scope and filter contract is retained.

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Slide210: GA4 Acquisition and Trend Contract — K5/A4 · Compare aligned periods and source scope; obtain competitor evidence separately.

Slide211: GA4 Engagement and Behaviour Indicators — A5/A6 · Report exact metric labels and period; benchmark only comparable populations and methods.

Slide212: GA4 Explorations: Variables to Technique — K7 · Explorations describe observed data; external models must keep assumptions and execution provenance.

Slide213: Forecast Model Comparison — K6/A7 · Compare predictions on a held-out period,then forecast; six months cannot resolve annual seasonality.

Slide214: Descriptive, Predictive and Causal Models — K7 · Distinguish descriptive,predictive,causal and allocation models before evaluating marketing effectiveness.

Slide215: Controlled Creative Experiment — A8 · Specify unit,control,primary metric,window and stop rule; significance needs a suitable statistical analysis.

Slide216: Conversion Rate, CPA and Revenue ROAS — K8 · Contribution after spend=2400×50%-1200=0; protect metric names and accounting boundaries.

Slide217: GA4 Key Events and Ads Conversions — LO4 · Current GA4 distinguishes key events from Ads conversions; a drafted contract is not live configuration evidence.

Slide218: Import and Join Integrity — Legacy data-import coverage retained as a data contract; no real customer identifiers are used in training.

Integrate supplement exhibits into the corresponding activity session;they do not add contact hours. Coverage includes primary questions,demographics,pricing,GA4 hierarchy,imports,report scope,explorations,forecast comparison and CPA/ROAS.

Resources and Delivery Controls

Current slide deck,Learner Guide,14 activity folders with mock CSVs,assets,prompt PDFs and guide PDFs;spreadsheet/Python3;authorised Cowork/Claude Code account where available. CSV fallback supports offline analytical practice. Do not expose trainer keys or use production customer data. Required admin slides precede teaching;embed brief introductions and attendance inside the opening session.

Assessment

Individual open book WA 60 minutes plus PP 90 minutes. Assessor observes K1–K8 and A1–A8;submit evidence on the matching LMS course. Minimum75% attendance and competent outcome;eligibility and funding terms apply. Flow:TRAQOM survey -> assessment digital attendance -> WA then PP -> LMS submission -> signed Assessment Summary Record.

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